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Delete a node from a binary search tree.

450. Delete Node in a BST Solved

Medium Topics Companies

Given a root node reference of a BST and a key, delete the node with the given key in the BST. Return the **root node reference** (possibly updated) of the BST.

Basically, the deletion can be divided into two stages:

1. Search for a node to remove.
2. If the node is found, delete the node.

Example 1:

```
class Solution {
    public TreeNode deleteNode(TreeNode root, int key) {
        if (root == null) return null;
        if (root.val == key) {
            root = deleteNode(root.left, key);
        } else if (root.val < key) {
            root.right = deleteNode(root.right, key);
        } else {
            if (root.left == null) return root.right;
            if (root.right == null) return root.left;
            else {
                TreeNode succ = findMin(root.right);
                root.val = succ.val;
                root.right = deleteNode(root.right, succ.val);
            }
        }
        return root;
    }
}
```

Check if a binary tree is a binary search tree.

98. Validate Binary Search Tree Solved

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Given the **root** of a binary tree, determine if it is a **valid binary search tree (BST)**.

A **valid BST** is defined as follows:

- The **left subtree** of a node contains only nodes with keys **strictly less than** the node's key.
- The **right subtree** of a node contains only nodes with keys **strictly greater than** the node's key.
- Both the **left** and **right** subtrees must also be **binary search trees**.

Example 1:

```
class Solution {
    public boolean isValidBST(TreeNode root) {
        return helper(root, Long.MIN_VALUE, Long.MAX_VALUE);
    }

    boolean helper(TreeNode node, long min, long max) {
        if (node == null) return true;
        if (node.val <= min || node.val >= max) return false;
        return helper(node.left, min, node.val) && helper(node.right, node.val, max);
    }
}
```

Find the kth smallest and kth largest elements in a BST.

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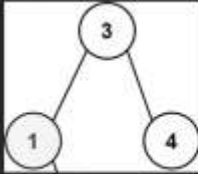
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230. Kth Smallest Element in a BST

Medium Topics Companies Hint

Given the `root` of a binary search tree, and an integer `k`, return the `k`th smallest value (1-indexed) of all the values of the nodes in the tree.

Example 1:



```

graph TD
    3((3)) --> 1((1))
    3 --> 4((4))
  
```

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Code

```

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24
16 class Solution {
17     int count=0;
18     int res=0;
19     public int kthSmallest(TreeNode root, int k) {
20         helper(root,k);
21         return res;
22     }
23     void helper(TreeNode root, int k){
24         if(root==null) return;
25         helper(root.left,k);
26         count++;
27         if(count==k){
28             res=root.val;
29             return;
30         }
31         helper(root.right,k);
32     }
33 }
  
```

Testcase Test Result

Find whether the given BST is balanced or not.

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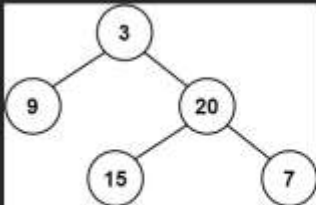
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110. Balanced Binary Tree

Easy Topics Companies

Given a binary tree, determine if it is **height-balanced**.

Example 1:



```

graph TD
    3((3)) --> 9((9))
    3 --> 20((20))
    20 --> 15((15))
    20 --> 7((7))
  
```

Inputs: root = [3,9,20,null,null,15,7]

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Code

```

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35
16 class Solution {
17     public boolean isBalanced(TreeNode root) {
18         if(root==null){
19             return true;
20         }
21         return helper(root)!=-1;
22     }
23     int helper(TreeNode root){
24         if(root==null){
25             return 0;
26         }
27         int left=helper(root.left);
28         if(left==-1){
29             return -1;
30         }
31         int right=helper(root.right);
32         if(right==-1){
33             return -1;
34         }
35         return 1;
36     }
37 }
  
```

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Testcase Test Result